

ShelfScape: Interactive Shelf Signage with Tangible Magnetic Attachments

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Figure 1: Shelf signage with magnet-attached price tag used for interaction detection.

Abstract

This paper presents ShelfScape, an interactive shelf signage system for enhancing customer engagement through tangible interactions. The system features a 90 cm-wide display with 64 inline Hall sensors to detect magnetic attachments behind the screen. Using this mechanism, four modules—Push, Slide, Rotate, and Hook—were designed to enable interactions via buttons, handles, plush toys, and repositionable tags. Each interaction triggers synchronized digital content. A lab study with 11 participants found the system left a strong impression, indicating its potential for novel in-store experiences.

CCS Concepts

• **Human-centered computing** → *Interaction devices*.

Keywords

interactive signage, tangible interface, retail technology

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1 Introduction

Digital signage technologies are widely used in urban, transportation, and retail settings, enabling personalized, real-time, and context-aware content delivery [Vogel and Balakrishnan 2004]. However, many systems suffer from *display blindness*—a phenomenon where users unconsciously ignore displays due to habituation or low relevance [Müller et al. 2009].

To address this, we focus on shelf signage: digital displays placed directly in front of merchandise. Located near price tags and within shoppers' natural line of sight, shelf signage is more likely to be seen. Yet conventional systems remain passive, typically showing static or looping content.

We developed a shelf signage system named ShelfScape that detects not only price tag positions but also physical states such as button presses and object movement via magnetic attachments. This enables a flexible platform for tangible input in retail environments. Using this system, we implemented four tangible interaction types—*push*, *slide*, *rotate*, and *hook*—by configuring magnetic attachments in different ways. Each module shares the same sensing mechanism and is demonstrated independently.

2 System Overview

We developed ShelfScape, a system that enables interactive content delivery through tangible user interactions with shelf-mounted signage. By detecting the position and type of magnetic attachments placed behind the screen, the system can respond to physical

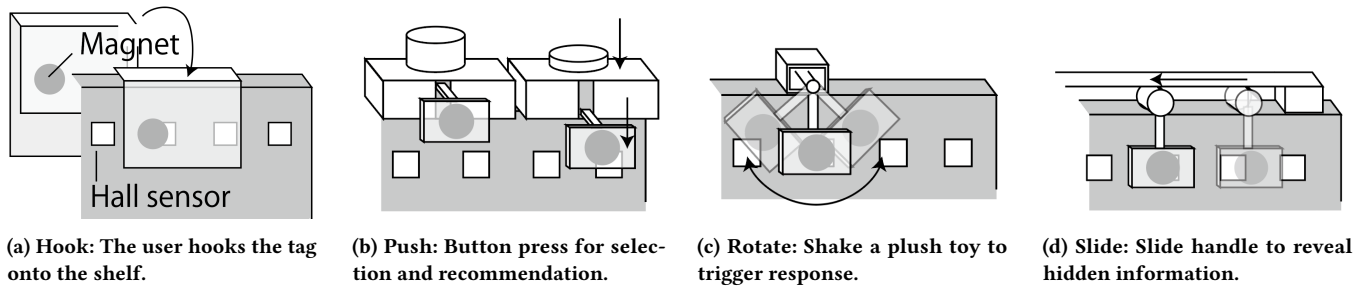


Figure 2: System overview and tangible interaction examples. All modules use the same magnetic sensing mechanism.



Figure 3: The number of magnets embedded in each tag allows the system to identify the product.

actions and present context-appropriate information and recommendations (Fig. 1).

ShelfScape consists of a 90cm-wide display unit (approx. 35 inches) embedded with 64 inline Hall effect sensors. These sensors are aligned horizontally to detect magnetic fields across the back surface of the display. The number of magnets embedded in each attachment enables identification of the module type (Fig. 3), allowing the system to recognize which item is placed where. This feature supports practical applications such as displaying targeted advertisements based on real product positions.

To explore the interactive potential of the system, we designed four modules that offer tangible and engaging experiences. Each module employs a custom attachment embedded with one or more magnets, placed on the back of the display and detected by the sensor array. Three signage units are installed on the shelf, and one interaction type is demonstrated at a time.

The **push interaction** uses button modules for quiz-like engagement. A question (e.g., “How are you feeling today?”) appears on the screen, with answer options and corresponding buttons below. Pressing a button moves the embedded magnet, which is detected by the system and triggers a product recommendation.

The **slide interaction** simulates a sliding door using a handle attachment. Sliding the handle shifts an internal magnet, sensed by the system. The on-screen graphic responds in sync, revealing hidden promotions.

The **rotate interaction** involves shaking or rotating a plush toy attachment (e.g., “Pet me!”). The motion causes the embedded magnet to shift position, which is detected and used to animate a character that suggests a product.

The **hook interaction** requires repositioning a fruit-themed tag that partially covers the screen. A character instructs the user to move the tag to a target location. When placed correctly, the embedded magnet aligns with a specific sensor position, triggering a character response and product suggestion. This mimics actions such as adjusting price tags or restocking.

3 Experiment

To evaluate the effectiveness of our shelf signage system, we conducted a laboratory study using four interactive content types: *Push*, *Slide*, *Rotate*, and *Hook*. These modules, showcasing various physical interactions via magnetic attachments, were presented sequentially on the same shelf display. We recruited 11 participants, who experienced all four interactions in a controlled setting. Afterward, they rated three items on a 5-point Likert scale: (Q1) willingness to use the system in an actual store, (Q2) memorability compared to ordinary digital signage, and (Q3) perceived usefulness. Average scores were 3.7 (Q1), 4.4 (Q2), and 3.3 (Q3).

The results suggest that participants saw potential for real-world use, especially when paired with engaging content. High memorability scores indicate that tangible interaction and visual feedback effectively captured attention. The lower usefulness score may reflect the lab setting, but participants still recognized value for product discovery. Several noted that animated character responses enhanced the experience, suggesting utility could increase in real retail environments.

4 Conclusion

This work presented a shelf signage system with magnetic attachments enabling tangible interactions such as *Push*, *Slide*, *Rotate*, and *Hook*. By transforming passive displays into interactive ones, the system enhances user engagement. A lab study showed improvements in attention and memorability. Future work includes real-world deployment to assess its effect on customer behavior and promotional impact.

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